



Diabetes 130-US Hospitals for Years 1999-2008

Donated on 5/2/2014

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The dataset represents ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks. Each row concerns hospital records of patients diagnosed with diabetes, who underwent...



Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification, Clustering

Feature Type

Categorical, Integer

Instances

101766

Features

47

Dataset Information

What do the instances in this dataset represent?

The instances represent hospitalized patient records diagnosed with diabetes.

Are there recommended data splits?

No recommendation. The standard train-test split could be used. Can use three-way holdout split (i.e., train-validation-test) when doing model selection.

Does the dataset contain data that might be considered sensitive in any way?

Citations/Acknowledgement

If you use this dataset, please cite:

Please cite:

Beata Strack, Jonathan P. DeShazo, Chris Gennings, Juan L. Olmo, Sebastian Ventura, Krzysztof J. Cios, and John N. Clore, "Impact ...



Creators

 **John Clore**

 **Krzysztof Cios**

Yes. The dataset contains information about the age, gender, and race of the patients.

Additional Information

The dataset represents ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks. It includes over 50 features representing patient and hospital outcomes. Information was extracted from the database for encounters that satisfied the following criteria....

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Has Missing Values?

Yes

 Jon DeShazo

 Beata Strack

DOI

10.24432/C5230J

License

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This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Introductory Paper

[Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Record](#)

By Beata Strack, Jonathan DeShazo, Chris Gennings, Juan Olmo, Sebastian Ventura, Krzysztof Cios, John Clore. 2014
Published in BioMed Research International, vol. 2014

Variables Table

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
encounter_id	ID			Unique identifier of an encounter		no
patient_nbr	ID			Unique identifier of a patient		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
race	Feature	Categorical	Race	Values: Caucasian, Asian, African American, Hispanic, and other		yes
gender	Feature	Categorical	Gender	Values: male, female, and unknown/invalid		no
age	Feature	Categorical	Age	Grouped in 10- year intervals: [0, 10), [10, 20),..., [90, 100)		no
weight	Feature	Categorical		Weight in pounds.		yes
admission_type_id	Feature	Categorical		Integer identifier corresponding to 9 distinct values, for example, emergency, urgent, elective, newborn, and not available		no
discharge_disposition_id	Feature	Categorical		Integer identifier corresponding to 29 distinct		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
				values, for example, discharged to home, expired, and not available		
admission_source_id	Feature	Categorical		Integer identifier corresponding to 21 distinct values, for example, physician referral, emergency room, and transfer from a hospital		no
time_in_hospital	Feature	Integer		Integer number of days between admission and discharge		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
payer_code	Feature	Categorical		Integer identifier corresponding to 23 distinct values, for example, Blue Cross/Blue Shield, Medicare, and self-pay		yes
medical_specialty	Feature	Categorical		Integer identifier of a specialty of the admitting physician, corresponding to 84 distinct values, for example, cardiology, internal medicine, family/general practice, and surgeon		yes
num_lab_procedures	Feature	Integer		Number of lab tests performed during the encounter		no
num_procedures	Feature	Integer		Number of procedures		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
				(other than lab tests) performed during the encounter		
num_medications	Feature	Integer		Number of distinct generic names administered during the encounter		no
number_outpatient	Feature	Integer		Number of outpatient visits of the patient in the year preceding the encounter		no
number_emergency	Feature	Integer		Number of emergency visits of the patient in the year preceding the encounter		no
number_inpatient	Feature	Integer		Number of inpatient visits of the patient in the year preceding the encounter		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
diag_1	Feature	Categorical		The primary diagnosis (coded as first three digits of ICD9); 848 distinct values		yes
diag_2	Feature	Categorical		Secondary diagnosis (coded as first three digits of ICD9); 923 distinct values		yes
diag_3	Feature	Categorical		Additional secondary diagnosis (coded as first three digits of ICD9); 954 distinct values		yes
number_diagnoses	Feature	Integer		Number of diagnoses entered to the system		no
max_glu_serum	Feature	Categorical		Indicates the range of the result or if the test was not taken. Values:		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
				>200, >300, normal, and none if not measured		
A1Cresult	Feature	Categorical		Indicates the range of the result or if the test was not taken. Values: >8 if the result was greater than 8%, >7 if the result was greater than 7% but less than 8%, normal if the result was less than 7%, and none if not measured.		no
metformin	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Val
				encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		

Rows per page

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Additional Variable Information



Detailed description of all the attributes is provided in Table 1 Beata Strack, Jonathan P. DeShazo, Chris Gennings, Juan L. Olmo, Sebastian Ventura, Krzysztof J. Cios, and John N. Clore, "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," ...

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Dataset Files



File	Size
diabetic_data.csv	18.3 MB

File	Size
IDS_mapping.csv	2.5 KB

Reviews

3.8



(3 ratings)

▲▼

LOGIN TO WRITE A REVIEW

Leland Burrill



3/30/2025

I fit an xgboost model w/ 0.96 accuracy and 0.99 AUC. I currently work with this kind of data. The trick here is to realize some patients have multiple records, and to engineer features that leverage that history, i.e, their past admissions. A histo...
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Michael Pluda



1/4/2025

Not a bad dataset, but definitely not for beginners or intermediates. This is a very imbalanced dataset, and even if you manage to soften this issue with under/oversampling techniques, the chance of distinguishing between the...
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Vincent Misch

a



how to solve the missing value of weight when data preprocessing ? cause there're way too much.

11/16/2024



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